Shaw Campbell

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Education

University of Cape Town — Cape Town, South Africa
BSc(Eng) Honours in Electrical & Computer Engineering — *2021 – 2025*

- Graduated with Honours; overall GPA 71%.
- Honours thesis: deep-learning deformable registration of ultrasound to MRI for prostate imaging (PyTorch, VoxelMorph, ANTs); semi-supervised training outperformed unsupervised, with affine pre-alignment and augmentation critical given limited data.

Experience

PSG Financial Services — Cape Town, South Africa
Graduate Software Developer (Intern) — *2026 – Present*

- Build and maintain production web applications (.NET / Angular, TypeScript) on an advice platform used by over one thousand financial advisors; write C# end-to-end tests with Playwright.
- Built agentic system (OpenCode, DeepSeek and AI Engineering concepts) to port dozens of automated tests from one language and framework to another, cutting the time taken for this task from months to weeks.

Peralex Electronics — Cape Town, South Africa
Software Engineering Intern — *2025 (5 weeks)*

- Designed and built a Python desktop app for visualising openEMS electromagnetic simulation results across multiple antenna types, within a full Agile cycle.

Electrum Payments — Cape Town, South Africa
Software Development Intern — *2024 (4 weeks)*

- Delivered a feature end-to-end on a live production codebase (Java, Spring Boot, REST APIs) embedded in the engineering team; received a full-time return offer.

University of Cape Town — Cape Town, South Africa
Tutor & Laboratory Assistant, Electrical Engineering & Computer Science — *2023 – 2025*

- Tutored Signals & Systems I/II and introductory Python/Java programming to classes of 300+ students; Control Systems Laboratory Assistant (2023, 2024).

Selected Projects

Multi-Modal Deformable Image Registration — 2026

- Converting messy thesis Jupyter notebooks into a clean, modular, documented Python repository for training image-registration models.

TaskFlow: CPU-Inspired Task Scheduler (C++) — 2025

- Translated a full UML design into a working task scheduler implementing classical CPU scheduling algorithms and OOP design patterns, with CMake.

Honey Badger Predator Deterrent System (EEE4114F) — 2024

- Developed the networking and UI for an ESP32-based predator deterrence system for a penguin conservation project with BirdLife South Africa.

Extracurricular & Awards

IT & Maintenance Representative, Leo Marquard Hall House Committee — 2023

- Served on a committee overseeing a residence of 400+ students and hundreds of thousands of rands in funds; helped secure winter heater funding for residents.

Deans Merit List Award — 2022 – 2023

UCT Plus Gold Award (for over 60 hours of elected leadership position) — 2023

PSG Financial Services Merit Based Bursary (Full-ride two year bursary) — 2024 – 2025

Skills

Languages: Python, Java, C++, C, C#, TypeScript/JavaScript, HTML/CSS
Tools & Frameworks: TensorFlow, .NET, ASP.NET, Angular, Spring Boot, Playwright, Jupyter, Git, Linux, MS SQL Server, NUnit, Selenium